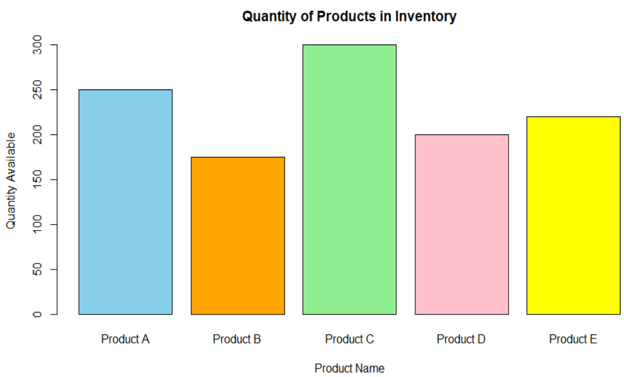
EXP-4

```r
# Product Inventory Dataset

inventory <- data.frame(
 Product_ID = c(1, 2, 3, 4, 5),
 Product_Name = c("Product A", "Product B", "Product C", "Product D", "Product E"),
 Quantity_Available = c(250, 175, 300, 200, 220)
)

# --------------------------------
# 1. Bar Chart - Quantity of Products
# --------------------------------

barplot(inventory$Quantity_Available,
    names.arg = inventory$Product_Name,
    col = c("skyblue", "orange", "lightgreen", "pink", "yellow"),
    main = "Quantity of Products in Inventory",
    xlab = "Product Name",
    ylab = "Quantity Available")
```



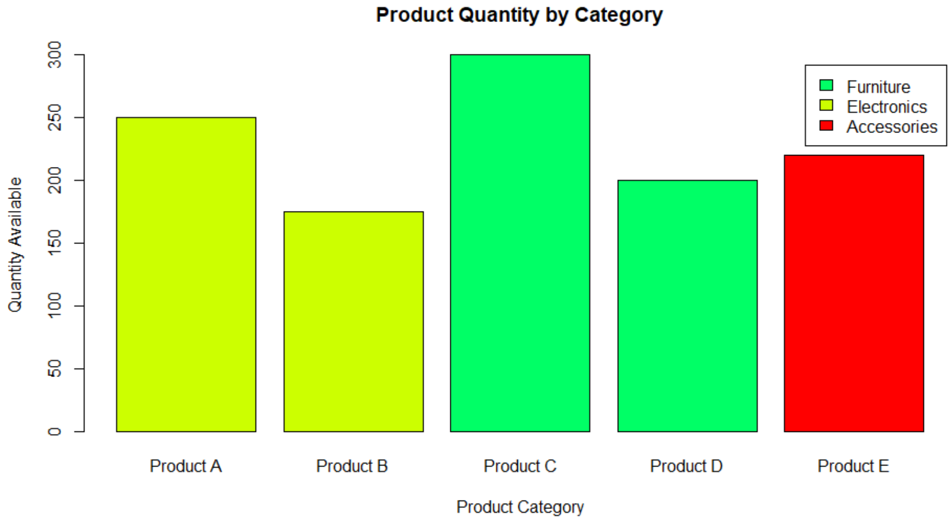

```r
# --------------------------------
# 2. Stacked Bar Chart
# --------------------------------
# Adding sample product categories

inventory$Category <- c("Electronics",
             "Electronics",
             "Furniture",
             "Furniture",
             "Accessories")

category_data <- xtabs(Quantity_Available ~ Category + Product_Name,
            data = inventory)

barplot(category_data,
    col = rainbow(5),
    main = "Product Quantity by Category",
```

```
        xlab = "Product Category",
        ylab = "Quantity Available",
        legend.text = rownames(category_data),
        beside = FALSE)
```




```
# --------------------------------
# 3. Scatter Plot - Price vs Quantity
# --------------------------------
# Sample product prices

inventory$Price <- c(500, 700, 450, 600, 550)

plot(inventory$Price,
     inventory$Quantity_Available,
     pch = 19,
     col = "red",
     xlab = "Product Price",
     ylab = "Quantity Available",
     main = "Product Price vs Quantity Available")

# Add regression line
abline(lm(Quantity_Available ~ Price, data = inventory),
       col = "blue",
       lwd = 2)

# Correlation
cor(inventory$Price,
    inventory$Quantity_Available)
```


Product Price vs Quantity Available
Quantity Available
Product Price
180
200
220
240
260
280
300
450
500
550
600
650
700